Parc-Fermé

FPGA-Based Data Acquisition System

Design, Component Selection and Verification Report

Simulation and generic synthesis assessment

Electrical and Electronics Subsystem

20 October 2026

Report status	Scope
Evidence-based draft; target FPGA and physical interfaces not finalized	Vehicle-level DAQ concept for rally and road-legal use; submitted technical evidence is limited to RTL simulation and generic Yosys synthesis.
Existing RTL evidence reviewed	rpm_period_measure.v and sampled_value_channel.v, their testbenches, logs, and the RPM waveform image available in the project evidence.

Prepared from the project repository and the current vehicle DAQ component shortlist.

Executive Summary

This report documents a simulation-first FPGA DAQ concept for seven vehicle measurements: engine RPM, coolant temperature, pre-turbo intake temperature, post-intercooler intake temperature, longitudinal acceleration, lateral acceleration and yaw rate. The vehicle BOM describes hardware that could be required in the actual car; the technical submission described here consists of RTL simulation and generic synthesis evidence. No physical vehicle integration or road/rally validation is claimed.

The supplied repository contains two RTL blocks: (1) rpm_period_measure, which synchronizes a digital pulse input and measures the number of FPGA clock cycles between rising edges, with reset and missing-pulse timeout handling; and (2) sampled_value_channel, a generic sampled-value register with a valid strobe. The supplied logs show both modules passing their included testbenches and being processed by Yosys without reported structural-check errors. This is useful module-level evidence, but it is not a complete seven-channel DAQ implementation.

Status category	Evidence-based status
Implemented and tested	RPM period measurement RTL and testbench; generic sampled-value capture RTL and testbench.
Synthesis evidence	Generic Yosys synthesis/check/stat logs for each of the two modules.
Planned system elements	CAN/IMU acquisition, temperature sensor/ADC or ECU-data interfaces, channel integration, timestamps/buffering and complete top-level design.
Not demonstrated	Target FPGA mapping, device-specific resource utilisation, timing closure, hardware bring-up, physical sensor calibration, vehicle CAN interoperability, automotive EMC/environmental validation.

1. Introduction

Data acquisition supports analysis of engine operation and vehicle dynamics by collecting time-dependent measurements. The proposed DAQ focuses on a deliberately limited set of seven channels rather than attempting to instrument every vehicle subsystem.

Channel	Measurement intent	Potential signal source / interface
1	Engine RPM	Existing ECU/tach output if documented and available; otherwise a suitable conditioned pulse source.

Channel	Measurement intent	Potential signal source / interface
2	Engine coolant temperature	Existing ECU value if available at the required measurement point; otherwise a compatible temperature sensor and analogue acquisition path.
3	Pre-turbo intake temperature	Measurement upstream of the turbocharger compressor; confirm actual sensor location and availability.
4	Post-intercooler intake temperature	Charge-air temperature measured downstream of the intercooler; confirm sensor location and availability.
5	Longitudinal acceleration	One axis of a six-axis accelerometer/gyroscope IMU.
6	Lateral acceleration	Second acceleration axis of the same IMU.
7	Yaw rate	Vertical-axis angular-rate measurement from the same IMU.

Scope boundary: the report describes a proposed vehicle-level architecture, but the current RTL and logs establish only the two modules listed above. The project README explicitly states that physical hardware integration and compatibility with actual vehicle sensors have not been verified.

2. Design Philosophy

- Simulation-first development: verify individual RTL blocks with repeatable testbenches before integration or hardware work.
- Incremental scope: preserve the completed RPM block and extend the design with clearly separated interfaces for the remaining measurements.
- Reuse before adding sensors: investigate whether the installed MoTeC M150 configuration already exposes the needed engine values and whether those values correspond to the exact required measurement locations.
- Off-the-shelf integration: avoid a custom PCB as a project constraint; use commercially available modules or mark an interface unresolved if no suitable module is available.
- Evidence-led claims: distinguish simulation, generic synthesis, vendor-device implementation and physical validation. Passing one stage does not imply the later stages passed.
- Vehicle suitability: component selection for an actual rally and road-legal car must consider electrical transients, vibration, temperature, moisture, connectors, mounting and serviceability—not price alone.

3. Problem Statement and Design Justification

The project seeks a compact DAQ concept that can gather a small, relevant set of engine and vehicle-dynamics parameters. The engineering problem is not only to represent the values in RTL; it is also to select a credible signal path from the physical measurement to the FPGA while avoiding unnecessary sensors and incompatible electrical interfaces.

The initial repository has intentionally open requirements. Its channel requirements file lists several other candidate measurements and marks their ranges/interfaces as TBD. For this report, the scope is restricted to the seven channels confirmed by the project owner; other candidate parameters are not included.

The immediate technical need is to establish a reusable RTL foundation and produce reproducible simulation and synthesis evidence. The vehicle-level design still needs the ECU signal list, exact sensor locations and part numbers, CAN protocol details, power requirements, and a finalized FPGA target before its physical bill of materials can be treated as purchase-ready.

4. System Architecture

Proposed end-to-end signal flow (conceptual; unimplemented blocks are explicitly shown as planned):

Stage	Proposed function	Status in the current repository
Physical source	ECU-provided engine data, temperature sensors, and a six-axis IMU	Not physically connected or verified.
Input/interface	Pulse conditioning for RPM; CAN receiver/controller for ECU/IMU if selected; ADC/conditioning for analogue temperature sensors if required	Not implemented in supplied RTL.
Measurement RTL	RPM period counter; generic sample capture; future channel-specific decoders/conversion	RPM period module and generic sampled-value module exist; full integration not present.
Data coordination	Common channel registers, sample-valid handling, timestamps, scaling and validity/error flags	Planned; not present as an integrated top-level design.
Verification	Self-checking testbenches, waveforms, simulation logs	Two module-level testbenches/logs are supplied.
Synthesis	Generic RTL synthesis/check/statistics; later device-specific implementation when target is selected	Generic Yosys reports supplied for the two modules only.

Interface note: the MoTeC M150 product family provides configurable input and CAN capabilities, but that does not prove the team's installed ECU publishes the exact requested signals. The installed ECU configuration, message identifiers, scaling, units, update rates and sensor locations must be checked with the powertrain team.

5. Decision Matrices and Component Justification

The following matrices are preliminary engineering decisions, not claims of completed validation. Weights prioritise vehicle compatibility and integration over low purchase price. Candidate prices are indicative supplier listings and must be reconfirmed before purchasing.

Criterion	Weight	Why it matters
Vehicle/environment suitability	30%	Rally vibration, thermal exposure, electrical noise and road use are more demanding than a lab bench.

Criterion	Weight	Why it matters
Interface compatibility	25%	Sensor output, logic voltage, CAN type/bitrate and physical connectors must match the architecture.
Measurement capability	20%	Range, accuracy, bandwidth, calibration and mounting determine usefulness of the data.
Cost and availability	15%	Keep the installed solution affordable and obtainable in India.
Integration / maintenance	10%	Prefer clear documentation, replaceable modules and serviceable wiring.

5.1 FPGA platform comparison

Candidate	Indicative listed price	Advantages	Limitations / decision
Altera MAX 10 EK-10M08E144	Rs. 5,073.71 (listing snapshot)	Lower-cost candidate; MAX 10 family includes an ADC feature, subject to exact device/board exposure.	Uses Quartus ecosystem; verify exact board manual and accessible pins/ADC. Development board is not automotive-qualified.
Digilent Cmod A7-35T (Artix-7)	Rs. 9,937.20 (listing snapshot)	Fits an AMD/Xilinx Vivado-oriented workflow; compact module and accessible digital I/O.	Higher cost; onboard analogue capability is limited and should not be assumed to cover all sensors. Not automotive-qualified.

Decision: no FPGA target is finalized in the current project. If the team prioritises lowest listed board price and accepts Quartus, the MAX 10 board is the cost-first candidate. If continuity with a Vivado-based workflow is more important, the Cmod A7 is a reasonable candidate. The existing Yosys reports are generic and do not select or prove either board.

5.2 Vehicle dynamics sensor comparison

Candidate	Strengths	Limitations	Decision
Bosch Motorsport MM7.10	Motorsport-oriented six-axis IMU; CAN output; published range and interface data.	Supplier quote, cable/connector variant, CAN message scaling and mounting details must be confirmed.	Preferred vehicle-grade candidate subject to budget and compatibility checks.
Low-cost hobby IMU breakout (e.g. MPU-6050 class)	Very low cost; useful for bench-level interface experiments.	Not equivalent to a calibrated motorsport sensor; board quality, vibration tolerance, environmental protection and vehicle suitability are not established.	Not recommended as the final installed rally/road sensor without a separate qualification case.

One six-axis IMU can provide longitudinal acceleration, lateral acceleration and yaw rate if its range, bandwidth, axes, calibration and CAN/data interface meet the requirements. A magnetometer is not required for these three specified channels.

5.3 Engine temperature and RPM signal strategy

Measurement	Preferred first option	Conditional fallback	Decision rationale
RPM	Documented ECU/tach output compatible with the existing pulse-period RTL.	Approved pulse conditioner/interface after source waveform and voltage are known.	Do not add a new crank sensor automatically; raw crank/cam signals must not be connected directly to FPGA I/O.
Coolant temperature	Existing ECU value at the correct sensor location, if available to the DAQ.	Bosch Motorsport NTC M12 candidate plus suitable analogue input/conditioning.	Reuse avoids duplication; exact thread, connector, resistance curve and engine fitment need confirmation.
Pre-turbo intake temperature	Existing signal only if its sensor is physically upstream of the compressor and its value is available.	Bosch Motorsport NTC M12-L candidate plus appropriate analogue interface.	Correct measurement location matters more than simply finding a generic intake-temperature sensor.
Post-intercooler intake temperature	Existing charge-air value only if measured downstream of the intercooler and available.	Second NTC M12-L candidate plus appropriate analogue interface.	Do not assume a turbo pressure sensor is also a temperature sensor without part-specific documentation.

5.4 CAN and power interface

The Digilent Pmod CAN 410-353 is a candidate off-the-shelf module for classical CAN 2.0B and SPI interfacing, but it must not be selected until the relevant network type, bitrate, connector topology and FPGA logic compatibility are confirmed. It is not a CAN-FD interface. The vehicle power front-end must be selected as a complete, suitably protected solution; a generic low-cost buck converter is not sufficient evidence of reverse-polarity, cranking-dip and load-dump protection.

6. Comparative Study

A single FPGA DAQ is attractive for deterministic, custom digital processing and flexible interfaces, but it is not automatically the least expensive complete vehicle logger. Reusing existing ECU measurements may reduce sensor and wiring costs; the trade-off is dependence on message configuration and bus access. Adding dedicated sensors gives measurement-location control but introduces sensor fitment, analogue conditioning, calibration and wiring work.

Approach	Advantages	Trade-offs	Fit for this project
Reuse existing ECU signals	Potentially lowest added cost; fewer duplicate sensors and less wiring.	Depends on ECU configuration, signal availability, update rate and scaling.	Preferred where signals are documented and located correctly.

Approach	Advantages	Trade-offs	Fit for this project
Add dedicated temperature sensors	Control over exact measurement location; independent measurement path.	Sensor fitment, analogue conditioning/ADC, calibration and extra wiring/cost.	Fallback if ECU values are unavailable or not at required locations.
CAN-output motorsport IMU	Single sensor supplies three requested dynamics channels; digital interface avoids three analogue channels.	Higher purchase cost; CAN mapping and mounting still need validation.	Preferred vehicle-level candidate if budget allows.
Hobby IMU breakout	Cheap and accessible for early experimentation.	Not proven equivalent for rally vibration/environment, accuracy or calibration.	Useful for bench testing only; not selected as final installed sensor.
Simulation-first RTL development	No hardware purchase required to demonstrate RTL behaviour; fast iteration and repeatable tests.	Cannot prove physical sensor compatibility, EMC, calibration or vehicle reliability.	Matches the current technical submission scope.

7. Simulation, Synthesis and Verification Particulars

7.1 Tools and evidence available

Tool / artifact	What the project evidence demonstrates
Icarus Verilog / vvp	Simulation logs for the RPM and sampled-value modules; reported checks pass.
GTKWave / VCD	An RPM waveform image and waves/rpm_period_measure.vcd are included.
Yosys 0.57+178	Generic synthesis/check/stat logs for both modules; each log reports zero structural problems in the Yosys check pass.
Target FPGA	Not finalized; no vendor-specific place-and-route or timing-closure evidence supplied.

7.2 RPM period measurement RTL

The rpm_period_measure module has a parameterized counter width (default 32) and timeout threshold (default 1000 clock cycles). It uses a two-stage synchronizer for the asynchronous rpm_pulse input, detects a rising edge, counts clock cycles between consecutive detected rising edges, asserts measurement_valid for one clock cycle when a period is captured, and asserts pulse_timeout if the threshold is reached without another pulse. Reset is active-low asynchronous.

Important limitation: this block outputs a pulse period in clock cycles; it does not itself convert the period to RPM units. Conversion requires the FPGA clock frequency and the number of input pulses per engine revolution, both of which must be specified and verified for the actual signal source.

RPM testbench scenario	Observed evidence in supplied log
Period capture	PASS: exact period count is 13.
Missing-pulse timeout	PASS: timeout detected.
Reset behaviour	PASS: reset clears measurement outputs.
Recovery after timeout/reset	PASS: recovery measurement has expected period count of 8.

RPM testbench scenario	Observed evidence in supplied log
Overall	ALL DIAGNOSTIC CHECKS PASSED.

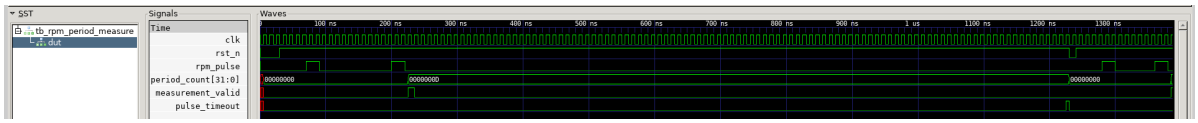


Figure 1. RPM period-measurement waveform evidence supplied in the project ZIP.

7.3 Generic sampled-value channel

The sampled_value_channel module is parameterized by DATA_WIDTH (16 in the included testbench). When sample_valid is asserted, it captures sample_in into sample_out and asserts measurement_valid. The supplied testbench checks values 95, 420 and 275, checks that valid clears after the capture pulse, and checks reset clears the output. The log reports all checks passed.

This is a generic register/valid-handshake block, not a temperature sensor interface, ADC driver, CAN decoder or IMU protocol implementation. It demonstrates a reusable capture primitive only.

Sampled-value test	Observed evidence
Capture sample 95	PASS
Capture sample 420	PASS
Capture sample 275	PASS
Reset clears outputs	PASS
Overall	ALL SAMPLED CHANNEL CHECKS PASSED.

7.4 Synthesis results

Module	Yosys generic cell statistics	Structural check
rpm_period_measure	25 cells: 1 add, 6 adff, 2 adffe, 1 and, 1 ge, 11 mux, 1 not, 1 reduce_and, 1 reduce_bool.	0 problems reported.
sampled_value_channel	3 cells: 1 adff, 1 adffe, 1 mux.	0 problems reported.

These are generic Yosys cell statistics, not counts of physical FPGA LUTs, flip-flops, DSPs, BRAMs or I/O resources. No target part was selected in the repository, and no device-specific timing report, clock constraint, place-and-route result or timing closure is included. Therefore the correct claim is that the two RTL modules were processed by generic Yosys synthesis and passed the logged structural checks—not that a complete FPGA DAQ has been mapped or timing-closed.

7.5 Verification coverage and gaps

Covered by supplied tests	Not covered in supplied ZIP
RPM period count for selected test intervals; timeout; reset; recovery.	Real sensor pulse waveform, pulse-per-revolution configuration, maximum RPM, jitter/noise/glitches, counter boundary cases, and physical input conditioning.
Sample capture; valid pulse clearing; reset.	Temperature conversion, ADC behaviour, CAN parsing/CRC/error frames, IMU axis/scaling, timestamps, multi-channel integration, buffer overflow and loss-of-data handling.

Covered by supplied tests	Not covered in supplied ZIP
Yosys structural check and generic statistics for each module.	Target-specific synthesis, timing constraints/closure, hardware testing, automotive EMC/environmental validation and calibrated measurement accuracy.

8. FMEA and Cost Analysis

8.1 Preliminary FMEA

Severity (S), occurrence (O) and detectability (D) are qualitative planning ratings from 1 (low/easy to detect) to 5 (high/difficult to detect). These are preliminary engineering judgements, not results of a formal safety analysis.

Failure mode	Effect	S/O/D	Mitigation / next action
RPM input voltage/waveform incompatible with FPGA	Damage, false edges or missed RPM periods.	5/3/4	Confirm source; use a suitable protected conditioner; define logic thresholds and maximum frequency; never connect raw crank/cam directly.
RPM pulse dropout/noise	Invalid period, timeout or erroneous RPM estimate.	3/3/2	Input synchronizer exists; add realistic noise/dropout tests and define timeout policy; use appropriate physical conditioning.
Incorrect pulses-per-revolution or clock frequency	Systematic RPM scaling error.	4/3/4	Document clock and pulse/revolution; implement and test conversion separately; compare against known reference.
Temperature signal missing/wrong location	Engine temperature reported incorrectly or not representative.	4/3/4	Verify sensor position and ECU channel mapping; add validity/range checks and calibration data.
CAN bitrate/message/scaling mismatch	No data or plausible but incorrect measurements.	4/3/3	Obtain DBC/message documentation; validate IDs, endianess, scale, offset, update rate and error handling.
IMU axes reversed or mounting misaligned	Wrong sign or cross-axis interpretation of vehicle dynamics.	3/3/3	Document coordinate convention; key and secure mounting; validate orientation and signs against known motions.

Failure mode	Effect	S/O/D	Mitigation / next action
Vehicle transient damages supply/interface	DAQ reset, corrupted data or component failure.	5/3/4	Select validated automotive input protection, fuse and regulator; review reverse polarity, cranking dips and load dump.
Vibration, moisture or connector disengagement	Intermittent measurements or total loss of DAQ data.	4/3/3	Retained locking connectors, strain relief, appropriate enclosure/glands, secure mounting and environmental review.
No device-specific timing closure	Unexpected timing failure on target FPGA.	3/2/3	Select target device, define clocks/constraints and run vendor implementation/timing analysis before hardware deployment.

8.2 Cost analysis (indicative, not a purchase quote)

The vehicle BOM is separate from the simulation/synthesis submission. The table below gives listed candidate prices captured during component research; these are not a fully costed installed-system total. Prices, tax treatment, freight and stock must be reconfirmed with vendors.

Item	Qty	Indicative unit price	Extended cost	Status / caveat
Digilent Cmod A7-35T FPGA development module	1	Rs. 9,937.20	Rs. 9,937.20	DigiKey listing snapshot; not automotive-qualified; target is not finalized.
Digilent Pmod CAN 410-353	1	Rs. 1,677.24	Rs. 1,677.24	Mouser listing snapshot; classical CAN only; use only if protocol/voltage compatibility confirmed.
AQUA AQ02 IP67 enclosure	1	Rs. 1,180.00	Rs. 1,180.00	Listed incl. GST; shipping extra; fit, heat and cable-entry sealing need checking.
Bosch Motorsport MM7.10 IMU	1	Quote required	Not priced	Motorsport candidate; exact order/cable variant and supplier quote required.
Coolant sensor, if needed	0–1	Quote required	Not priced	Conditional on ECU signal availability and correct fitment.

Item	Qty	Indicative unit price	Extended cost	Status / caveat
Pre-turbo IAT sensor, if needed	0–1	Quote required	Not priced	Conditional on ECU signal availability and correct measurement location.
Post-intercooler IAT sensor, if needed	0–1	Quote required	Not priced	Conditional on ECU signal availability and correct measurement location.
Protected automotive power interface, connectors and wiring	As designed	Not priced	Not priced	Must be selected for actual load, transient protection, harness route and installation.

Subtotal of the three publicly priced candidates above: Rs. 12,794.44 before shipping and any additional components. This is not the complete vehicle DAQ cost, and it should not be used as a final budget until the quote-required and conditional lines are resolved. The team may also avoid buying an FPGA board if an appropriate board is already available.

9. Conclusion and References

The supplied project demonstrates early, modular RTL work rather than a complete vehicle DAQ. The RPM period-measurement module and generic sampled-value channel each have testbenches, passing simulation logs and generic Yosys synthesis/check/stat reports. The current evidence supports a simulation-first milestone and provides a starting point for future channel integration.

The recommended next engineering actions are: (1) confirm the exact seven-channel requirements and measurement locations; (2) obtain the installed M150 CAN/message configuration and identify which engine values are available; (3) confirm the RPM source waveform and pulse-per-revolution; (4) finalize the FPGA target and interface voltage requirements; (5) obtain an IMU quote and confirm its CAN map; (6) select a properly protected vehicle power solution; (7) implement and verify the missing ADC/temperature/CAN/IMU interfaces; and (8) run target-specific synthesis/timing analysis once the device is selected.

References and datasheets

R1. Project repository supplied by the project owner: README.md; requirements/channel_requirements.md; docs/architecture.md; docs/decisions.md; components/component_selection.md; docs/references.md; RTL, testbenches and reports/evidence logs (ZIP attachment).

R2. Yosys Open SYnthesis Suite, version 0.57+178 as reported in the supplied logs. https://yosyshq.net/yosys/

R3. Icarus Verilog project documentation. https://steveicarus.github.io/iverilog/

R4. MoTeC M150 product information. General product capabilities must not be interpreted as proof of the installed ECU's configured message list. https://www.motec.com.au/products/M150

R5. Digilent Cmod A7-35T product page / board reference. https://digilent.com/shop/cmod-a7-35t-breadboardable-artix-7-fpga-module/

R6. Digilent Pmod CAN product documentation (410-353). Confirm the exact revision and current datasheet before purchase. https://digilent.com/shop/pmod-can-can-2-0b-controller-with-integrated-transceiver/

R7. Bosch Motorsport MM7.10 inertial measurement unit, manufacturer specifications. https://www.bosch-motorsport.com/products/sensors/vehicle-dynamics/inertial-measurement-unit-mm7-10/

R8. Bosch Motorsport NTC M12 temperature sensor, manufacturer specifications. https://www.bosch-motorsport.com/products/sensors/temperature/temperature-sensor-ntc-m12/

R9. Bosch Motorsport NTC M12-L temperature sensor, manufacturer specifications. https://www.bosch-motorsport.com/products/sensors/temperature/temperature-sensor-ntc-m12-l/

R10. Texas Instruments TIDA-01179 automotive input transient/cold-crank reference design. This is a design reference, not necessarily a retail finished module. https://www.ti.com/tool/TIDA-01179

R11. AQUA AQ02 IP67 enclosure supplier listing; verify current dimensions, price and assembly instructions. https://vkenterprises.tech/product/aqua-series-ip67-waterproof-enclosure-for-electronics-aq02/

Appendix A — Exact evidence filenames in supplied ZIP

- rtl/rpm_period_measure.v
- tb/tb_rpm_period_measure.v
- reports/evidence/rpm_simulation.log
- reports/evidence/rpm_yosys.log
- reports/evidence/rpm_period_measurement.png
- waves/rpm_period_measure.vcd
- rtl/sampled_value_channel.v
- tb/tb_sampled_value_channel.v
- reports/evidence/sampled_value_simulation.log
- reports/evidence/sampled_value_yosys.log

Evidence note: the project evidence also contains compiled .vvp simulation artifacts. This report bases claims on the RTL, testbench source, logs and waveform file; it does not interpret the presence of a compiled artifact as evidence of hardware deployment.